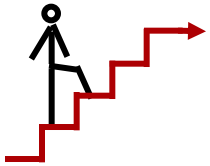
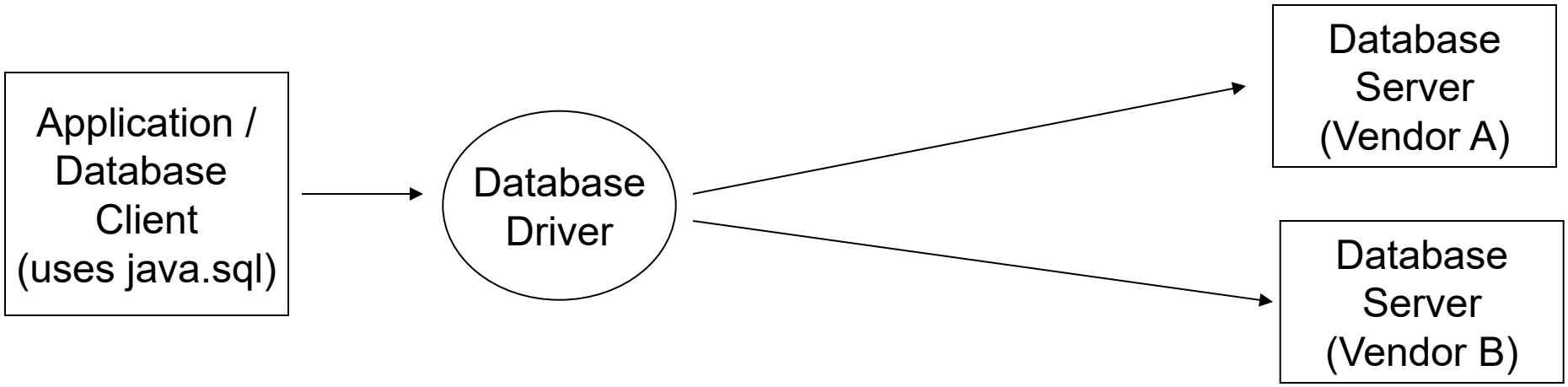


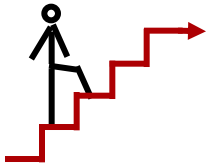
Working with Databases using JDBC



JDBC BASICS

- **JDBC usage can be segregated into following :**
  1. **Application – that uses java.sql API to retrieve/query a database.**
  2. **Database – a repository for data organized in a structured way.**
  3. **Database Driver – A separate entity which provides interface between the Application and Database.**

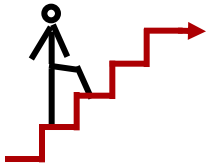




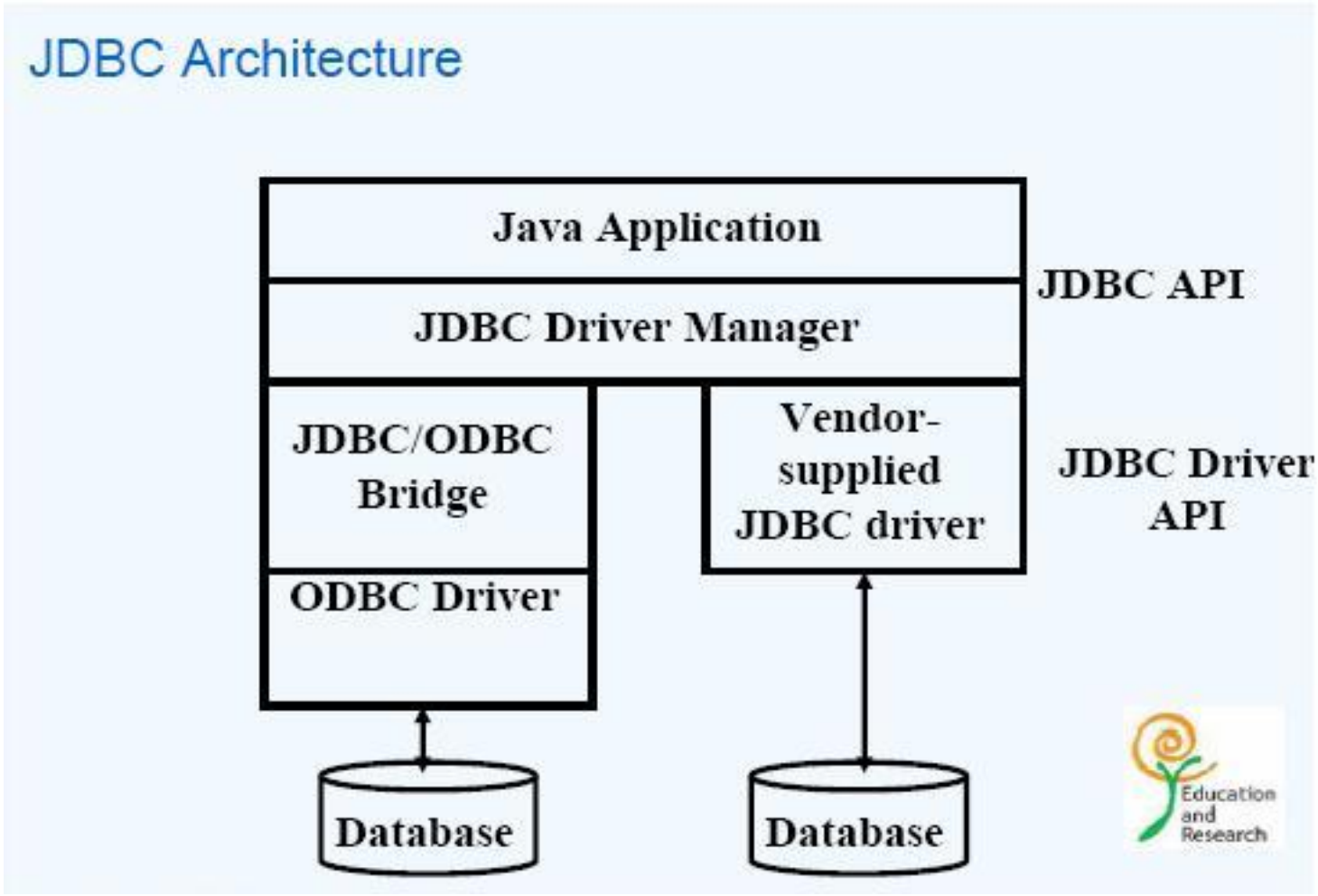
JDBC

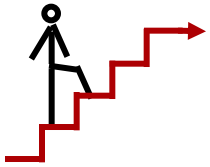
JDBC is a Java API for executing SQL statements **(A single program will be able to send SQL statements to the appropriate database)**

The API consists of classes and interfaces to send SQL statements to different database **(JDBC is a low level API that supports basic SQL functionality)**

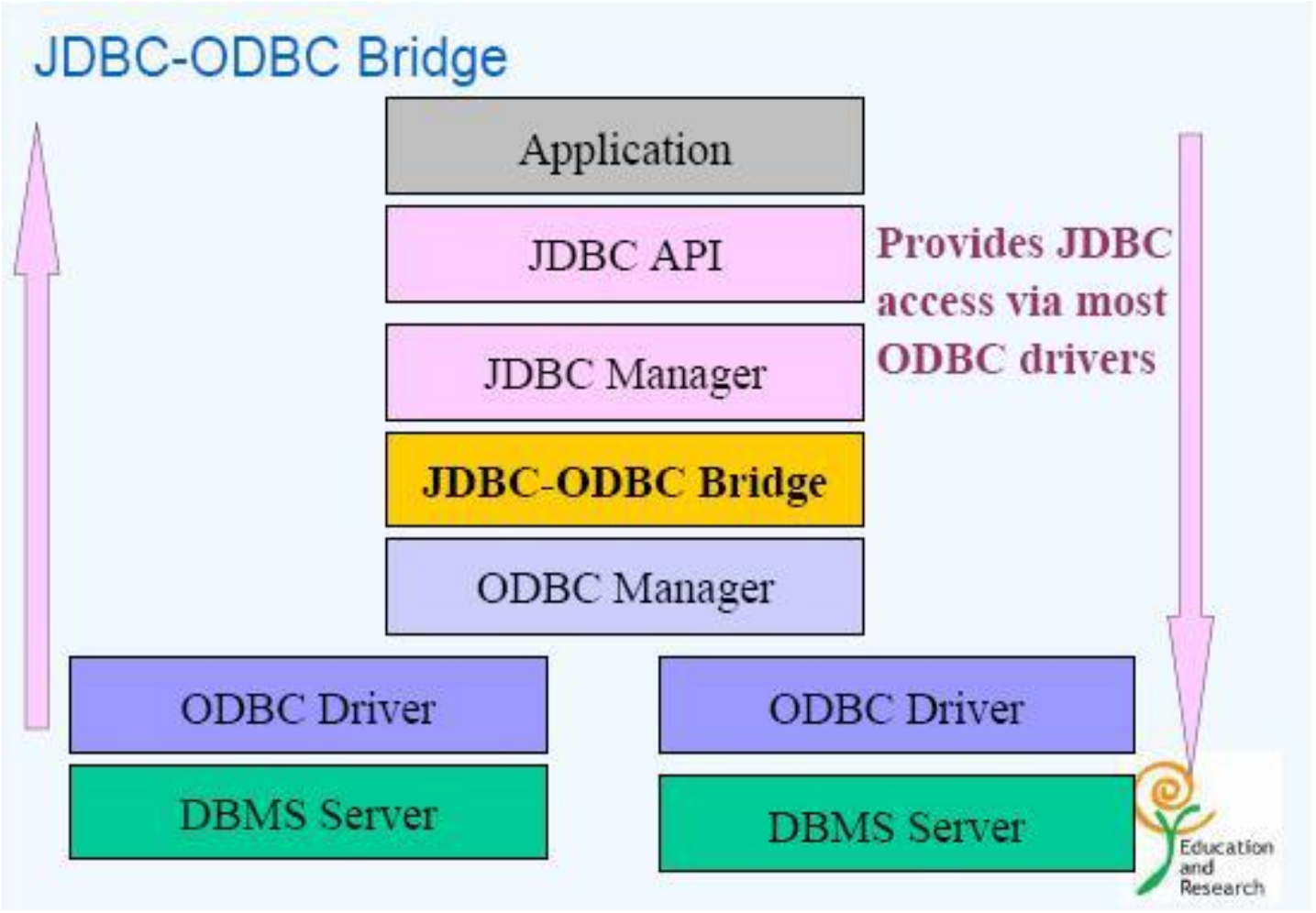


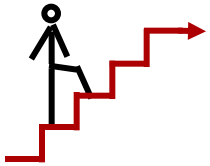
JDBC - Architecture





JDBC-ODBC Bridge





JDBC-ODBC Bridge

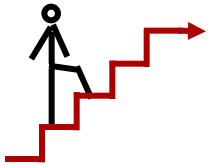
- **To create this database, you can feed SQL statements to an ODBC data source via the JDBC-ODBC bridge.**

  First, you will have to create an ODBC data source.

  You must load a driver that tells the JDBC classes how to talk to a data source. In this case, you will need the class JdbcOdbcDriver:

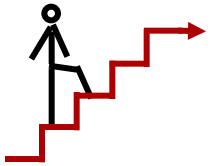
  Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");

  This can also be specified from the command line via the jdbc.drivers system property:



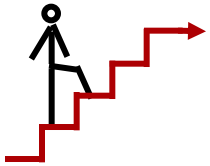
Steps to using JDBC driver

1. Load the database driver
2. Establish a connection to the database
3. Create a statement object
4. Execute SQL Query statement
5. Retrieve the ResultSet Object
6. Retrieve the records/field data from ResultSet object for processing
7. Close ResultSet Object
8. Close Statement Object
9. Close Connection Object



Using java.sql

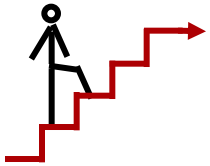
- **Class DriverManager**
  - **Allows access to information about drivers and setting up of connections**
    - public Connection getConnection(String url) → Protocol:subprotocol:name
    - public Connection getConnection(String url, String name, String password)
    - public Connection getConnection(String url, Properties args)
    - public Driver getDriver(String url)
    - public Enumeration getDrivers()
    - public void registerDriver(Driver)
    - public void deregisterDriver(Driver)
    - public void setLoginTimeout(int)
    - public int getLoginTimeout()
    - public void setLogStream(PrintStream)



Using java.sql

- **Interface Connection**
  - **An object that implements the interface Connection encapsulates a connection to database.**

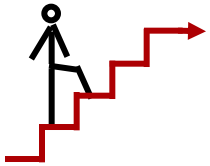
public Statement	createStatement()
public PreparedStatement	prepareStatement(String)
public CallableStatement	prepareCall(String)
public String	nativeSQL(String)
public void	setAutoCommit(boolean)
public boolean	getAutoCommit()
public void	commit()
public void	rollback()
public void	close()
public boolean	isClosed()
public DatabaseMetaData	getMetaData()
public SQLWarning	getWarnings()
public void	clearWarnings()
public Savepoint	setSavepoint()
public void	rollback(Savepoint)
public void	releaseSavepoint(Savepoint)



Using java.sql

- **Interface DatabaseMetaData**
  - **Gives information about structure and capabilities of a database.**

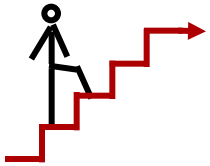
```
public boolean          allProceduresAreCallable()
public boolean          allTablesAreSelectable()
public java.lang.String getURL()
public java.lang.String getUserName()
public boolean          isReadOnly()
public java.lang.String getDatabaseProductName()
public java.lang.String getDatabaseProductVersion()
public java.lang.String getDriverName()
public java.lang.String getDriverVersion()
public int              getDriverMajorVersion();
public int              getDriverMinorVersion();
public boolean          usesLocalFiles()
public boolean          supportsMixedCaseIdentifiers()
```



Using java.sql

- **Interface DatabaseMetaData (Contd.)**

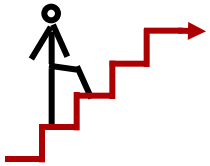
```
public boolean      storesUpperCaseIdentifiers()
public boolean      storesLowerCaseIdentifiers()
public boolean      storesMixedCaseIdentifiers()
public String       getSQLKeywords()
public boolean      supportsGroupBy()
public boolean      supportsMultipleResultSets()
public boolean      supportsMultipleTransactions()
public boolean      supportsNonNullableColumns()
```



Using java.sql

- **Interface Statement**
  - **Provides a way to perform SQL queries**

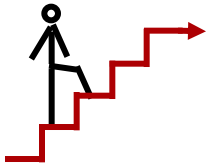
```
public boolean          execute(java.lang.String)
public java.sql.ResultSet          executeQuery(java.lang.String)
public int              executeUpdate(java.lang.String)
public void         close()
public int          getMaxFieldSize()
public void         setMaxFieldSize(int)
public int          getMaxRows()
public void         setMaxRows(int)
public int          getQueryTimeout()
public void         setQueryTimeout(int)
public java.sql.SQLWarning        getWarnings()
public void                       clearWarnings()
```



Using java.sql

- **Interface ResultSet**
  - **It encapsulates the records retrieved from a query statement**

public java.lang.String	getString(int)
public boolean	getBoolean(int)
public byte	getByte(int)
public short	getShort(int)
public int	getInt(int)
public long	getLong(int)
public float	getFloat(int)
public double	getDouble(int)
public java.sql.Date	getDate(int)
public java.sql.Time	getTime(int)
public java.sql.Timestamp	getTimestamp(int)

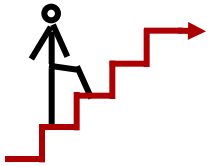


Using java.sql

- **Interface ResultSet (Contd.)**

```
public String                getString(java.lang.String)
public boolean               getBoolean(java.lang.String)
public byte                  getByte(java.lang.String)
public short                 getShort(java.lang.String)
public int                   getInt(java.lang.String)
public long                  getLong(java.lang.String)
public float                 getFloat(java.lang.String)
public double                getDouble(java.lang.String)
public java.sql.Date         getDate(java.lang.String)
public java.sql.Time         getTime(java.lang.String)
public java.sql.Timestamp          getTimestamp(java.lang.String)

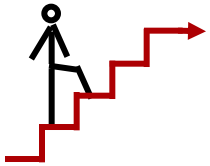
public boolean               next()
public void                  close()
public java.sql.SQLWarning         getWarnings()
public void                  clearWarnings()
public java.sql.ResultSetMetaData          getMetaData()
```



Using java.sql

- **Interface ResultSetMetaData**
  - **It provides information about contents of a ResultSet**

public java.lang.String	getTableName(int)
public boolean	isReadOnly(int)
public boolean	isWritable(int)
public int	getColumnCount()
public java.lang.String	getColumnTypeName(int)
public int	getColumnDisplaySize(int)
public java.lang.String	getColumnLabel(int)
public java.lang.String	getColumnName(int)
public java.lang.String	getSchemaName(int)
public int	getColumnType(int)
public boolean	isCurrency(int)
public int	isNullable(int)
public boolean	isSigned(int)



Using java.sql

- **Interface PreparedStatement**
  - **Provides a way to use precompiled SQL statements**

public java.sql.ResultSet executeQuery()
public boolean execute()
public int executeUpdate()
public void setNull(int, int)
public void setBoolean(int, boolean)
public void setByte(int, byte)
public void setShort(int, short)
public void setInt(int, int)
public void setLong(int, long)
public void setFloat(int, float)
public void setDouble(int, double)
public void setString(int, java.lang.String)
public void setDate(int, java.sql.Date)
public void setTime(int, java.sql.Time)
public void clearParameters()
public java.sql.ResultSetMetaData getMetaData()
public java.sql.ParameterMetaData getParameterMetaData()